

The Fourier Decay of Rvachev's Up-Function

A final resolution of the pointwise, metric, averaged, and extremal regimes

Prepared for Vladimir Reshetnikov's ProveIt development

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Abstract

Let

$$f(x) = \widehat{\text{up}}(x) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z},$$

where the Fourier convention is $\widehat{h}(\xi) = \int_{\mathbb{R}} h(t) e^{-2\pi i t \xi} dt$. The motivating question asks for the real-axis decay of f , and in particular for a positive analytic, eventually decreasing gauge g for which the ordinary Cesàro mean of $|f|/g$ either converges to one or remains bounded between two positive constants. Four first-wave reports and a later comparative audit produced several apparently competing answers.

The competition disappears once the word “amplitude” is separated into its inequivalent meanings. The exact identity

$$|f(2^k(1+t))| = \frac{2^{-k(k+1)/2}}{\pi^{k+1}(1+t)^{k+1}} H(1+t) \prod_{j=0}^k |\sin(\pi 2^j t)|, \quad 0 \leq t \leq 1,$$

reduces the problem to a lacunary product for the doubling map. Every natural size has the common log-normal core

$$\exp\left(-\frac{(\log x)^2}{2 \log 2}\right),$$

but its power of x depends on whether one takes a geometric mean, an arithmetic mean, an RMS mean, or a supremum. The respective exponents are

$$\begin{aligned} \kappa_0 &= 3.151496129472319\dots, & \kappa_1 &= 2.748070014871334\dots, \\ \kappa_2 &= 2.651496129472318\dots, & \kappa_{\infty} &= 2.359014879111741\dots \end{aligned}$$

The arithmetic exponent is governed by the Perron root $\rho_1 = 0.661322602060565\dots$ of an explicit transfer operator. With

$$E_{\kappa}(x) = x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right)$$

and $A_1 = 0.0912661241315\dots$, the elementary gauge $g = A_1 E_{\kappa_1}$ is positive, real analytic, eventually decreasing, and satisfies the requested two-sided Cesàro bound. The normalized mean tends to one at dyadic endpoints. For unrestricted $t = 2^k y$, however, it converges to a nonconstant dyadic-mantissa profile determined by a singular conformal eigenmeasure. The same obstruction defeats every gauge whose within-shell shape stabilizes, including the usual elementary, regularly varying, and continuous log-periodic corrections. This is the precise negative result; the materials do not prove nonexistence among arbitrary scale-dependent analytic gauges. A logarithmic Cesàro average, with measure dx/x , does converge for all t after the explicit rescaling $A_1^{\log} = 0.0938606357568\dots$

For almost every fixed dyadic ray the centered logarithm satisfies a CLT and a two-sided law of the iterated logarithm with asymptotic variance $\pi^2/4$ per dyadic step. This corrects a factor 2 in the variance calculation in the Aistleitner–Hofer–Larcher source: the sharp coefficient in front of $\sqrt{n \log \log n}$ is $\pi/\sqrt{2}$, not π .

Finally, if $M_k = \max_{2^k \leq x \leq 2^{k+1}} |f(x)|$, put $a = \log 2$, $\beta = \log(\sqrt{3}/2)$, $c = (a/\pi)\sqrt{3/2}$, and $n = k + 1$. Then

$$\log M_k = -\frac{a}{2}k(k+1) - n \log \pi + n\beta - \frac{W(cn)^2 + 2W(cn)}{2a} + \mathcal{O}((\log \log k)^2).$$

Thus the true annular envelope contains a second log-normal suppression in $\log \log x$, not a fixed power of $\log x$. Individual interzero peaks are still more irregular: immediately after $m2^k$, with m fixed and odd, they obey an explicit much smaller two-adic asymptotic. Together these results give a single consistent resolution of all supplied materials.

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1 The question, the conventions, and the final verdict

Rvachev's up-function is an even, nonnegative C^∞ function supported on $[-1, 1]$ and normalized to have integral one. Under the Fourier convention in the abstract, its transform is the infinite sinc product above; see Arias de Reyna [2], the Fabius-function summary [13], and the formal and expository development in the ProveIt repository [10].

The original question [9] asks for a positive analytic, eventually decreasing g such that, for a sufficiently large t_0 ,

$$\frac{1}{t - t_0} \int_{t_0}^t \frac{|f(x)|}{g(x)} dx \longrightarrow 1, \quad t \rightarrow \infty, \quad (1)$$

or at least such that this expression stays between two positive constants. The latter condition will be denoted by Q3; the full limit will be denoted by Q2.

A single pointwise equivalent cannot exist: f vanishes at every nonzero integer. More importantly, even after removing those zeros, four different operations on the same lacunary numerator give four different exponential rates. Put

$$a := \log 2, \quad E_\kappa(x) := x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2a}\right). \quad (2)$$

The complete first-order spectrum is as follows.

Notion of size	numerator rate Λ	exponent $\kappa = \frac{\log \pi + a/2 - \log \Lambda}{a}$	conclusion
Geometric mean / a.e. fixed ray	$1/2$	$\kappa_0 = 3.1514961295 \dots$	Typical logarithmic law, with CLT and LIL fluctuations.
Arithmetic mean (L^1)	$\rho_1 = 0.6613226021 \dots$	$\kappa_1 = 2.7480700149 \dots$	Correct exponent for the Cesàro question; Q3 holds.
Root mean square (L^2)	$1/\sqrt{2}$	$\kappa_2 = \log_2(2\pi) = 2.6514961295 \dots$	Exact identity $\int_0^1 P_n^2 = 2^{-n}$.
Uniform fixed-ray envelope	$\sqrt{3}/2$	$\kappa_\infty = 2.3590148791 \dots$	Sharp global upper envelope, attained on the $1/3 \leftrightarrow 2/3$ orbit.

The exponent $\log_2(2\pi)$ proposed in the early draft is therefore not a random near miss. It is exactly the RMS exponent. It is not the pointwise, arithmetic-mean, or supremum exponent, and no fixed power of $\log x$ can repair a mismatch already proportional to $\log x$ in the logarithm.

Final answer to the original normalization question. Let

$$g_1(x) = A_1 E_{\kappa_1}(x), \quad A_1 = 0.0912661241315 \dots$$

Then Q3 holds for all sufficiently large t . The Q2 limit equals one along $t = 2^k$, but for $t = 2^k y$, $1 \leq y \leq 2$, it converges to a nonconstant continuous profile $\mathcal{P}(y)$. Hence this natural elementary gauge does not satisfy unrestricted Q2. No gauge with a stabilized continuous within-shell shape can satisfy Q2. This includes all standard log-normal power gauges, slowly varying corrections, and continuous log-periodic modulations. The unrestricted class of arbitrary scale-dependent analytic monotone gauges is not eliminated by the supplied arguments and should not be claimed to be eliminated.

1.1 Division between elementary proof, imported analysis, and numerics

The exact product identities, the zero and lobe structure, the L^2 identity, the Gelfond bound, the Lambert- W optimization, the singularity argument, and the two-adic peak asymptotic are proved below directly.

The metric results use standard ergodic and martingale limit theorems. The finite- q shell limits use the standard Ruelle–Perron–Frobenius spectral theorem for a contracting two-branch weighted transfer operator. The existence of the L^1 exponential rate also follows from Fouvry–Mauduit [6]; the rigorous published enclosure $0.66130 < \rho_1 < 0.66135$ is due to Aistleitner–Hofer–Larcher [1]. Decimal digits beyond a certified enclosure are numerical, not substitutes for proof.

2 Entire product, zero multiplicities, signs, and lobes

2.1 Convergence, scaling, and the canonical zero product

The infinite product

$$f(z) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^j}\right) \quad (3)$$

converges locally uniformly on $\mathbb{C}$. Indeed, $\log \operatorname{sinc} w = -w^2/6 + \mathcal{O}(w^4)$ near zero, and $\sum_j |z|^2/4^j$ is locally uniformly bounded. Hence f is even and entire, $f(0) = 1$, and its factors may be regrouped locally uniformly.

Proposition 2.1 (Three exact product descriptions). *For every $z \in \mathbb{C}$,*

$$f(2z) = \operatorname{sinc}(2\pi z) f(z), \quad (4)$$

$$f(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}, \quad (5)$$

$$f(z) = \operatorname{sinc}(\pi z) H(z), \quad H(z) := \prod_{j=1}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^j}\right). \quad (6)$$

Thus every nonzero integer m is a zero of multiplicity $1 + \nu_2(|m|)$, and there are no other zeros.

Proof. Shifting the index in (3) gives (4); separating the first factor gives (6). Euler's product

$$\operatorname{sinc}(\pi z) = \prod_{r=1}^{\infty} \left(1 - \frac{z^2}{r^2}\right)$$

applied at $z/2^j$ shows that a fixed integer m appears once for each $j \in \{0, \dots, \nu_2(m)\}$. This gives (5) and the stated zero multiplicities. $\square$

2.2 Exactly one extremum in every interzero interval

The Weierstrass product exposes more than the zeros. Away from the integers, termwise logarithmic differentiation gives

$$(\log |f|)'(x) = -2x \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^2 - x^2}, \quad (7)$$

$$(\log |f|)''(x) = -2 \sum_{m=1}^{\infty} (1 + \nu_2(m)) \frac{m^2 + x^2}{(m^2 - x^2)^2} < 0. \quad (8)$$

The differentiated series converge locally uniformly off the zero set.

Theorem 2.2 (Lobe structure and signed Thue–Morse pattern). *The function f decreases strictly from 1 to 0 on $(0, 1)$. For every integer $m \geq 1$, the interval $(m, m+1)$ contains exactly one critical point c_m ; it is the unique maximum of $|f|$ on that interval. Moreover,*

$$\operatorname{sgn} f(x) = (-1)^{s_2(m)}, \quad m < x < m+1, \quad (9)$$

where $s_2(m)$ is the sum of the binary digits of m .

Proof. Strict negativity in (8) makes $(\log |f|)'$ strictly decreasing on every interzero interval. On $(m, m+1)$ its pole at the left endpoint has limit $+\infty$, while its pole at the right endpoint has limit $-\infty$; hence it has exactly one zero. On $(0, 1)$ it starts at zero and is strictly decreasing, so f is strictly decreasing there.

Crossing the zero at r changes the sign $1 + \nu_2(r)$ times. The sign on $(m, m+1)$ is therefore

$$(-1)^{\sum_{r=1}^m (1 + \nu_2(r))}.$$

Legendre's identity $\nu_2(m!) = m - s_2(m)$ gives $\sum_{r=1}^m (1 + \nu_2(r)) = 2m - s_2(m)$, whose parity is $s_2(m)$. $\square$

The theorem legitimizes speaking of “the peak” in each unit interval, but it does not make the peak sequence regular. In [section 9](#) we will see that the peaks immediately following large powers of two are extraordinarily deep troughs.

3 The exact dyadic reduction

Put

$$s(t) := |\sin(\pi t)|, \quad \mathbb{T}t := 2t \pmod{1}, \quad P_n(t) := \prod_{j=0}^{n-1} s(\mathbb{T}^j t). \quad (10)$$

The product P_n is, up to 2^{-n} , the modulus of the length- 2^n Thue–Morse trigonometric polynomial:

$$2^n P_n(t) = \left| \prod_{j=0}^{n-1} (1 - e^{2\pi i 2^j t}) \right| = \left| \sum_{r=0}^{2^n-1} (-1)^{s_2(r)} e^{2\pi i r t} \right|. \quad (11)$$

Theorem 3.1 (Exact shell factorization). *Let $k \geq 0$, $n = k + 1$, $0 \leq t \leq 1$, and $y = 1 + t$. Then*

$$\boxed{|f(2^k y)| = 2^{-k(k+1)/2} (\pi y)^{-n} H(y) P_n(t).} \quad (12)$$

Here H is given by (6); it is positive on $[1, 2)$ and vanishes at 2.

Proof. Iterating (4) gives

$$f(2^k y) = f(y) \prod_{j=1}^k \operatorname{sinc}(\pi 2^j y).$$

Since 2^j is an integer, $|\sin(\pi 2^j y)| = |\sin(\pi 2^j t)|$. The k new denominators multiply to $\pi^k y^k 2^{k(k+1)/2}$, while $|f(y)| = H(y) |\sin(\pi t)| / (\pi y)$. Collecting the $n = k + 1$ sine factors proves the formula. $\square$

3.1 One algebraic normalization for all notions of size

For a candidate numerator growth rate $0 < \Lambda < 1$, define

$$\kappa(\Lambda) := \frac{\log \pi + a/2 - \log \Lambda}{a} \quad (13)$$

and

$$W_\Lambda(y) := H(y) \frac{\Lambda}{\pi} y^{\kappa(\Lambda)-1} \exp\left(\frac{(\log y)^2}{2a}\right), \quad 1 \leq y \leq 2. \quad (14)$$

The function W_Λ is continuous, nonnegative, bounded, and positive on $[1, 2)$.

Corollary 3.2 (Master normalization identity). *With the notation of [theorem 3.1](#),*

$$\boxed{\frac{|f(2^k y)|}{E_{\kappa(\Lambda)}(2^k y)} = \Lambda^{-n} W_\Lambda(y) P_n(t).} \quad (15)$$

Proof. Write $b = \log y$ and $\log(2^k y) = ka + b$. The logarithm of the deterministic factor in (12), divided by E_κ , is

$$k(\kappa a - \log \pi - a/2) + \frac{b^2}{2a} + (\kappa - 1)b - \log \pi.$$

For (13), the coefficient of k is $-\log \Lambda$. Moving one further factor Λ^{-1} into the bounded mantissa term gives exactly (14) and (15). $\square$

The triangular denominator $2^{-k(k+1)/2}$ is responsible for the universal quadratic term in $\log x$. Every disagreement in the lower-order term is now localized in the operation applied to P_n .

4 Typical dyadic rays and the corrected metric fluctuation law

4.1 Birkhoff averages and the typical exponent

Set

$$\psi(t) := \log |2 \sin(\pi t)| = a + \log s(t). \quad (16)$$

This function belongs to $L^p(0, 1)$ for every finite p , and

$$\int_0^1 \psi(t) dt = 0. \quad (17)$$

For $S_n(t) := \sum_{j=0}^{n-1} \psi(T^j t)$, the master identity with $\Lambda = 1/2$ becomes the exact equality

$$\log |f(2^k(1+t))| + \frac{(\log(2^k(1+t)))^2}{2a} + \kappa_0 \log(2^k(1+t)) = \log W_{1/2}(1+t) + S_n(t), \quad (18)$$

where

$$\kappa_0 = \kappa(1/2) = \frac{3}{2} + \frac{\log \pi}{a} = 3.151496129472319 \dots \quad (19)$$

Theorem 4.1 (Typical ray law). *For Lebesgue-almost every $t \in (0, 1)$,*

$$\log |f(2^k(1+t))| = -\frac{(\log(2^k(1+t)))^2}{2a} - \kappa_0 \log(2^k(1+t)) + o(k). \quad (20)$$

Proof. The doubling map is ergodic for Lebesgue measure. Birkhoff's theorem applied to the integrable centered function (16) gives $S_n(t) = o(n)$ almost everywhere. Substitute this into (18). $\square$

More generally, whenever the orbit average $\alpha(t) = \lim_{n \rightarrow \infty} n^{-1} \sum_{j < n} \log s(T^j t)$ exists and is finite, the same algebra gives the ray exponent

$$\kappa(\alpha) = \frac{\log \pi + a/2 - \alpha}{a}. \quad (21)$$

Thus arithmetic properties of the binary expansion remain visible on exceptional rays.

4.2 An exact reverse-martingale decomposition

Let P be the Perron operator of the doubling map,

$$(Ph)(x) = \frac{1}{2} (h(x/2) + h((x+1)/2)). \quad (22)$$

The double-angle identity gives

$$P\psi = \frac{1}{2}\psi. \quad (23)$$

Define

$$d := 2\psi - \psi \circ \mathbb{T}. \quad (24)$$

Then

$$d(x) = \log |\tan(\pi x)|, \quad Pd = 0, \quad (25)$$

and therefore $d \circ \mathbb{T}^j$ is a stationary reverse martingale-difference sequence. Telescoping gives

$$S_n(t) = \sum_{j=0}^{n-1} d(\mathbb{T}^j t) - \psi(t) + \psi(\mathbb{T}^n t). \quad (26)$$

Lemma 4.2 (Exact asymptotic variance). *One has*

$$\int_0^1 d(x)^2 dx = \frac{\pi^2}{4}. \quad (27)$$

Equivalently,

$$\int_0^1 \psi(x) \psi(\mathbb{T}^r x) dx = \frac{\pi^2}{12 \cdot 2^r}, \quad r \geq 0, \quad (28)$$

and hence

$$\text{Var}(S_n) = \frac{\pi^2}{12} (3n - 4 + 2^{2-n}) = \frac{\pi^2}{4} n + \mathcal{O}(1). \quad (29)$$

Proof. By symmetry and the standard beta-integral derivatives,

$$\int_0^1 \log^2 |\tan(\pi x)| dx = \frac{2}{\pi} \int_0^{\pi/2} \log^2(\tan u) du = \frac{\pi^2}{4}.$$

For the Fourier proof, use

$$\psi(x) = - \sum_{m=1}^{\infty} \frac{\cos(2\pi m x)}{m} \quad \text{in } L^2(0, 1). \quad (30)$$

Orthogonality contributes the factor $1/2$ and gives

$$\int \psi(x) \psi(2^r x) dx = \frac{1}{2} \sum_{m=1}^{\infty} \frac{1}{(2^r m)m} = \frac{\pi^2}{12 \cdot 2^r}.$$

Summing the covariance series yields (29). $\square$

Theorem 4.3 (CLT and two-sided LIL). *If t is Lebesgue distributed on $(0, 1)$, then*

$$\frac{S_n(t)}{(\pi/2)\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (31)$$

For Lebesgue-almost every fixed t ,

$$\limsup_{n \rightarrow \infty} \frac{S_n(t)}{(\pi/2)\sqrt{2n \log \log n}} = 1, \quad (32)$$

$$\liminf_{n \rightarrow \infty} \frac{S_n(t)}{(\pi/2)\sqrt{2n \log \log n}} = -1. \quad (33)$$

Proof. The variable d has finite moments of every order. Since $Pd = 0$, the martingale CLT and martingale LIL apply to the first term on the right of (26), with variance (27); see Hall–Heyde [8] and Stout [12]. The logarithmic singularities of ψ have exponentially decreasing tails, so $\psi(\mathbb{T}^n t) = o(\sqrt{n \log \log n})$ almost surely by Borel–Cantelli. The fixed term $\psi(t)$ is harmless. $\square$

In the original frequency variable, the almost-sure upper and lower log-amplitude excursions have size

$$\pm \frac{\pi}{\sqrt{2a}} \sqrt{\log x \log \log x} \quad \text{along a typical dyadic ray.} \quad (34)$$

They are larger than every fixed multiple of $\log \log x$, so no factor $(\log x)^p$ can make the typical normalized amplitude converge.

4.3 Correction to the published lacunary-product constant

Aistleitner–Hofer–Larcher [1] state, in their Theorem 2, a coefficient π in front of $\sqrt{n \log \log n}$ for the upper excursions of $\prod_{j < n} |2 \sin(\pi 2^j t)|$. Their displayed variance calculation writes

$$\sum_{m \geq 1} \left(\frac{1}{m^2} + 2 \sum_{r \geq 1} \frac{1}{2^r m^2} \right) = \frac{\pi^2}{2}.$$

The cosine orthogonality factor $1/2$ is missing. Formula (28) shows that the correct Green–Kubo variance is

$$\frac{1}{2} \sum_{m \geq 1} \frac{1}{m^2} + 2 \sum_{r \geq 1} \frac{1}{2} \sum_{m \geq 1} \frac{1}{2^r m^2} = \frac{\pi^2}{4}. \quad (35)$$

Consequently the correct coefficient relative to $\sqrt{n \log \log n}$ is

$$\boxed{\frac{\pi}{\sqrt{2}}}, \quad (36)$$

not π . The exact martingale decomposition above supplies an independent proof and also yields the missing lower LIL. The third first-wave report inherited the published coefficient; the first report and the comparative audit correctly detected the variance issue.

5 The complete finite- L^q shell spectrum

5.1 The weighted doubling operator

For $q > 0$, define

$$(\mathcal{L}_q h)(x) := \frac{1}{2} \left[\sin^q \left(\frac{\pi x}{2} \right) h(x/2) + \cos^q \left(\frac{\pi x}{2} \right) h((x+1)/2) \right]. \quad (37)$$

This is the Perron operator of the doubling map with weight s^q . A change of variables on the two inverse branches gives the exact duality formula

$$\int_0^1 h(t) P_n(t)^q dt = \int_0^1 \mathcal{L}_q^n h(x) dx. \quad (38)$$

The inverse branches contract by $1/2$, and the weights are Hölder of every order not exceeding $\min(1, q)$. The endpoint zeros do not destroy primitivity: at $x = 0$ and $x = 1$, the surviving branch has positive weight. The standard Ruelle–Perron–Frobenius theorem therefore gives the following; see Baladi [3] or Ruelle [11].

Proposition 5.1 (Spectral data). *For each $q > 0$, the operator $\mathcal{L}_q$ on a suitable Hölder space has a simple leading eigenvalue $\rho_q > 0$, a strictly positive eigenfunction h_q , and a positive eigenmeasure ν_q . They may be normalized by*

$$\mathcal{L}_q h_q = \rho_q h_q, \quad \mathcal{L}_q^* \nu_q = \rho_q \nu_q, \quad \nu_q(h_q) = 1. \quad (39)$$

There is $0 < \vartheta_q < 1$ such that

$$\rho_q^{-n} \mathcal{L}_q^n h = h_q \nu_q(h) + \mathcal{O}(\vartheta_q^n \|h\|_{C^\alpha}) \quad (40)$$

for compatible Hölder h .

Set

$$\Lambda_q := \rho_q^{1/q}, \quad \kappa_q := \kappa(\Lambda_q) = \frac{\log \pi + a/2 - \log \Lambda_q}{a}. \quad (41)$$

For a function F on the k th shell, write

$$\mathcal{M}_{q,k}(F) := \left(2^{-k} \int_{2^k}^{2^{k+1}} |F(x)|^q dx \right)^{1/q}. \quad (42)$$

Theorem 5.2 (Sharp finite- q shell normalization). *For every $q \in (0, \infty)$ there is $A_q \in (0, \infty)$ such that*

$$\mathcal{M}_{q,k} \left(\frac{f}{E_{\kappa_q}} \right) \longrightarrow A_q, \quad k \rightarrow \infty. \quad (43)$$

More precisely,

$$A_q^q = \left(\int_0^1 h_q(x) dx \right) \nu_q(W_{\Lambda_q}(1 + \cdot)^q). \quad (44)$$

The quantity Λ_q is nondecreasing in q , and κ_q is nonincreasing.

Proof. Use $x = 2^k(1+t)$ in (42) and raise (15) to the q th power. Since $\Lambda_q^{-nq} = \rho_q^{-n}$,

$$\mathcal{M}_{q,k} \left(\frac{f}{E_{\kappa_q}} \right)^q = \rho_q^{-n} \int_0^1 W_{\Lambda_q}(1+t)^q P_n(t)^q dt.$$

Apply (38) and (40). This gives (43) and (44). For each fixed n , the L^q norm of P_n on the probability space $[0, 1]$ is nondecreasing in q . Take n th roots and then $n \rightarrow \infty$ to obtain monotonicity of Λ_q ; formula (41) reverses the order for κ_q . $\square$

A factor $(\log x)^p$ is not part of any sharp finite- q normalization. At the correct κ_q , inserting such a factor multiplies the normalized shell mean by $(k \log 2)^{-p}(1 + o(1))$; it tends to zero for $p > 0$ and infinity for $p < 0$.

5.2 The four distinguished values

Geometric mean, $q = 0$. Because Lebesgue measure is invariant under $\mathbb{T}$ and $\int_0^1 \log s = -a$, the limiting numerator rate is $1/2$. In fact, the normalized shell geometric mean at κ_0 is exactly independent of k :

$$\exp \left(2^{-k} \int_{2^k}^{2^{k+1}} \log \frac{|f(x)|}{E_{\kappa_0}(x)} dx \right) = \exp \left(\int_0^1 \log W_{1/2}(1+t) dt \right). \quad (45)$$

This is the shell form of [theorem 4.1](#).

Arithmetic mean, $q = 1$. Write

$$\rho := \rho_1. \quad (46)$$

Fouvry and Mauduit proved

$$\int_0^1 P_n(t) dt = K \rho^n (1 + o(1)), \quad K > 0, \quad (47)$$

and Aistleitner–Hofer–Larcher established

$$0.66130 < \rho < 0.66135. \quad (48)$$

High-precision transfer-operator computation gives

$$\rho = 0.661322602060565 \dots \quad (49)$$

Thus

$$\kappa_1 = \frac{1}{2} + \frac{\log(\pi/\rho)}{a} = 2.748070014871334 \dots \quad (50)$$

This is the exponent relevant to the ordinary absolute mean.

Root mean square, $q = 2$. Since

$$\mathcal{L}_2 \mathbf{1} = \frac{1}{2} \left(\sin^2 \frac{\pi x}{2} + \cos^2 \frac{\pi x}{2} \right) = \frac{1}{2} \mathbf{1}, \quad (51)$$

we have $\rho_2 = 1/2$ and

$$\int_0^1 P_n(t)^2 dt = 2^{-n} \quad (52)$$

for every n . Consequently

$$\kappa_2 = \log_2(2\pi) = 2.651496129472318 \dots \quad (53)$$

This identifies precisely what the original draft computed.

Uniform norm, $q = \infty$. The limit is $\Lambda_\infty = \sqrt{3}/2$ and gives

$$\kappa_\infty = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{a} = 2.359014879111741 \dots \quad (54)$$

A direct sharp proof is given in [section 7](#).

The strict ordering

$$\kappa_\infty < \kappa_2 < \kappa_1 < \kappa_0 \quad (55)$$

is the cleanest summary of why the four reports appeared to disagree.

6 The ordinary Cesàro question

6.1 The sharp arithmetic gauge and Q3

Let

$$W_1(y) := W_\rho(y) = H(y) \frac{\rho}{\pi} y^{\kappa_1-1} \exp\left(\frac{(\log y)^2}{2a}\right). \quad (56)$$

Define

$$A_1 := \left(\int_0^1 h_1(x) dx \right) \nu_1(W_1(1 + \cdot)) = 0.0912661241315 \dots \quad (57)$$

By [theorem 5.2](#),

$$2^{-k} \int_{2^k}^{2^{k+1}} \frac{|f(x)|}{E_{\kappa_1}(x)} dx \longrightarrow A_1. \quad (58)$$

Theorem 6.1 (Answer to Q3 and dyadic Q2). *Put*

$$g_1(x) := A_1 E_{\kappa_1}(x). \quad (59)$$

For every fixed $t_0 > 0$, there are $0 < c < C < \infty$ such that

$$c \leq \frac{1}{t - t_0} \int_{t_0}^t \frac{|f(x)|}{g_1(x)} dx \leq C \quad (60)$$

for all sufficiently large t . Moreover,

$$\frac{1}{2^K - t_0} \int_{t_0}^{2^K} \frac{|f(x)|}{g_1(x)} dx \longrightarrow 1. \quad (61)$$

Proof. Let J_k be the integral of $|f|/E_{\kappa_1}$ over $[2^k, 2^{k+1}]$. Equation (58) says $J_k = A_1 2^k (1 + o(1))$. Summing the geometric sequence of completed shells proves (61). If $2^K \leq t < 2^{K+1}$, the preceding complete shell already contributes a fixed positive multiple of 2^K , while the sum of all completed shells and the whole current shell is at most a fixed multiple of 2^K . Since $t \asymp 2^K$, this proves (60). $\square$

6.2 Analyticity and eventual derivative signs

The requested regularity is not an issue for (59).

Proposition 6.2 (Eventual alternating derivatives). *The function g_1 is positive and real analytic on $(0, \infty)$. For each fixed integer $m \geq 0$, there exists X_m such that*

$$(-1)^m g_1^{(m)}(x) > 0, \quad x \geq X_m. \quad (62)$$

In particular it is eventually strictly decreasing, and every fixed derivative is eventually monotone.

Proof. Put $u(x) = \log x/a + \kappa_1$. Then $g_1'(x) = -x^{-1}u(x)g_1(x)$. Induction using $d/dx = x^{-1}d/d \log x$ gives

$$\frac{g_1^{(m)}(x)}{g_1(x)} = x^{-m} \left((-1)^m u(x)^m + \mathcal{O}_m(u(x)^{m-1}) \right). \quad (63)$$

The leading term controls the sign for sufficiently large x . $\square$

The threshold X_m depends on m ; this is not a claim of complete monotonicity on one common half-line.

6.3 The nonconstant dyadic-mantissa profile

The shell limit is stronger than mere boundedness but weaker than unrestricted Q2. Normalize h_1, ν_1 as in (39) and define

$$\mathcal{F}(z) := \frac{\nu_1(W_1(1 + \cdot)\mathbf{1}_{[0,z]})}{\nu_1(W_1(1 + \cdot))}, \quad 0 \leq z \leq 1. \quad (64)$$

Lemma 6.3 (The conformal measure is atomless and singular). *The eigenmeasure ν_1 has no atoms and is singular with respect to Lebesgue measure. The same is true after multiplication by the continuous weight $W_1(1 + z)$.*

Proof. For an atom at z , the eigenmeasure equation gives

$$\rho \nu_1(\{z\}) = \frac{1}{2} s(z) \nu_1(\{Tz\}). \quad (65)$$

Endpoint atoms vanish because $s = 0$. If a remaining atom existed, its forward masses would grow by at least the factor $2\rho > 1$ until an orbit repeated; a periodic orbit gives an immediate contradiction and a nonperiodic orbit gives infinitely many atoms of unbounded mass. Hence ν_1 is atomless.

Suppose its absolutely continuous component has a nonzero density $r \in L^1$. The Lebesgue decomposition is preserved by the two nonsingular inverse branches, so that component separately satisfies

$$\rho r(x) = s(x)r(Tx) \quad \text{for a.e. } x. \quad (66)$$

Iteration gives

$$r(T^n x) = \frac{\rho^n r(x)}{P_n(x)}. \quad (67)$$

For Lebesgue-a.e. x , Birkhoff's theorem gives $P_n(x) = 2^{-n+o(n)}$, so the right side grows like $(2\rho)^{n+o(n)}r(x)$. On the other hand, for every $\varepsilon > 0$, Markov's inequality and Borel–Cantelli imply $r(T^n x) \leq e^{\varepsilon n}$ eventually for a.e. x , because T preserves Lebesgue measure. Taking $0 < \varepsilon < \log(2\rho)$ is a contradiction on the set $r > 0$. Thus ν_1 is singular. Multiplying by a continuous positive weight away from a single atomless endpoint preserves singularity and absence of atoms. $\square$

Because the weighted measure is atomless, $\mathcal{F}$ is continuous.

Theorem 6.4 (Full dyadic phase limit). *For every $y \in [1, 2]$ and every fixed $t_0 > 0$,*

$$\lim_{K \rightarrow \infty} \frac{1}{2^K y - t_0} \int_{t_0}^{2^K y} \frac{|f(x)|}{A_1 E_{\kappa_1}(x)} dx = \mathcal{P}(y) := \frac{1 + \mathcal{F}(y - 1)}{y}. \quad (68)$$

The function $\mathcal{P}$ is continuous, $\mathcal{P}(1) = \mathcal{P}(2) = 1$, and it is not identically one. Consequently g_1 does not satisfy unrestricted Q2.

Proof. From (15) and $x = 2^K(1 + t)$,

$$2^{-K} \int_{2^K}^{2^{K(1+z)}} \frac{|f(x)|}{E_{\kappa_1}(x)} dx = \rho^{-n} \int_0^z W_1(1 + t) P_n(t) dt. \quad (69)$$

The spectral decomposition (40), followed by approximation of $\mathbf{1}_{[0,z]}$ from above and below by Hölder functions, shows that the right side tends to $A_1 \mathcal{F}(z)$. All preceding complete shells contribute $A_1 2^K(1 + o(1))$. Division by $A_1 2^K y$ proves (68).

If $\mathcal{P} \equiv 1$, then $1 + \mathcal{F}(y - 1) = y$, or $\mathcal{F}(z) = z$. This would make the normalized weighted eigenmeasure Lebesgue measure, contradicting theorem 6.3. $\square$

6.4 Exactly what kind of Q2 gauge is ruled out

The singular profile yields a robust impossibility theorem, but its hypotheses matter.

Theorem 6.5 (No stabilized within-shell gauge). *Suppose g is positive for large x and that there exist positive constants c_K and a positive continuous function u on $[0, 1]$ such that*

$$\frac{g(2^K(1 + z))}{c_K E_{\kappa_1}(2^K(1 + z))} \rightarrow u(z) \quad (70)$$

uniformly for $0 \leq z \leq 1$. Then g cannot satisfy Q2.

Proof. Under (70), the normalized partial-shell measures are, up to $1/c_K + o(1)$,

$$\rho^{-n} \frac{W_1(1 + z)}{u(z)} P_n(z) dz.$$

The spectral theorem makes these converge, after total-mass normalization, to the probability measure proportional to $W_1(1 + z)u(z)^{-1} d\nu_1(z)$. Q2 would force every partial shell to have asymptotic mass equal to its Euclidean length, hence would force that limiting measure to be Lebesgue. It is singular by theorem 6.3, a contradiction. $\square$

Slowly varying corrections satisfy (70) with $u = 1$. A continuous log-periodic multiplier gives a continuous positive u . The same applies to the standard elementary and Hardy-field log-normal gauges once the correct quadratic and linear terms have been fixed. What the theorem does *not* address is a deliberately scale-dependent analytic gauge whose shape develops ever finer structure from shell to shell. The comparative audit's phrase "unattainable in principle" is therefore too broad if read as a theorem about every conceivable analytic monotone function.

6.5 A logarithmic Cesàro mean removes the phase

The phase survives ordinary averaging because the newest dyadic shell has the same order as all preceding shells combined. With the measure dx/x , every shell has equal logarithmic length and the final incomplete shell is negligible.

Define

$$B_1 := \lim_{k \rightarrow \infty} \int_{2^k}^{2^{k+1}} \frac{|f(x)|}{E_{\kappa_1}(x)} \frac{dx}{x} = 0.0650592350404 \dots \quad (71)$$

and

$$A_1^{\log} := \frac{B_1}{a} = 0.0938606357568 \dots \quad (72)$$

The existence of the limit follows from the same spectral argument, with $W_1(1+t)$ replaced by $W_1(1+t)/(1+t)$.

Corollary 6.6 (All-phase logarithmic Cesàro limit). *For every fixed $t_0 > 0$,*

$$\frac{1}{\log(t/t_0)} \int_{t_0}^t \frac{|f(x)|}{A_1^{\log} E_{\kappa_1}(x)} \frac{dx}{x} \rightarrow 1. \quad (73)$$

Proof. Every complete dyadic shell contributes $B_1 + o(1)$ to the unscaled numerator, whereas its logarithmic length is a . The number of complete shells is asymptotic to $\log t/a$. The one incomplete shell contributes only $\mathcal{O}(1)$ and is negligible after division by $\log t$. $\square$

7 The sharp fixed-ray and uniform envelope

The L^∞ numerator rate can be proved by one elementary inequality. Put

$$\lambda := \frac{\sqrt{3}}{2}, \quad \beta := \log \lambda. \quad (74)$$

Lemma 7.1 (Sharp two-step sine inequality). *For every real x ,*

$$\frac{2}{3} \log s(x) + \frac{1}{3} \log s(\mathbb{T}x) \leq \log \lambda, \quad (75)$$

with the usual value $-\infty$ at a zero. Equality holds exactly at $x \equiv 1/3$ or $2/3 \pmod{1}$.

Proof. Cubing the exponential form reduces the assertion to $s(x)^2 s(\mathbb{T}x) \leq \lambda^3$. Write $u = \sin^2(\pi x)$. The square of the left side is $4u^3(1-u)$, whose maximum on $[0, 1]$ is $27/64$, attained only at $u = 3/4$. Taking square roots gives $3\sqrt{3}/8 = \lambda^3$ and identifies the equality orbit. $\square$

Proposition 7.2 (Exact exponential rate of the finite supremum). *For every $n \geq 1$,*

$$\lambda^n \leq \|P_n\|_\infty \leq \lambda^{n-1}. \quad (76)$$

Hence

$$\lim_{n \rightarrow \infty} \|P_n\|_\infty^{1/n} = \lambda. \quad (77)$$

Proof. At $x = 1/3$, the orbit alternates between $1/3$ and $2/3$, so every factor is λ . For the upper bound, sum (75) over $x, \mathbb{T}x, \dots, \mathbb{T}^{n-2}x$. If $\phi = \log s$, the result is

$$2\phi(x) + 3 \sum_{j=1}^{n-2} \phi(\mathbb{T}^j x) + \phi(\mathbb{T}^{n-1} x) \leq 3(n-1) \log \lambda.$$

Adding the nonpositive boundary combination $\phi(x) + 2\phi(\mathbb{T}^{n-1}x)$ shows $3 \sum_{j < n} \phi(\mathbb{T}^j x) \leq 3(n-1) \log \lambda$. Exponentiate. $\square$

This is the classical Gelfond exponent for the Thue–Morse polynomial; the more general ergodic-optimization setting is discussed by Fan, Schmeling, and Shen [5].

Theorem 7.3 (Sharp global envelope). *With κ_∞ from (54), there is $C < \infty$ such that*

$$|f(x)| \leq CE_{\kappa_\infty}(x), \quad x \geq 1. \quad (78)$$

The exponent is sharp. More precisely, for every $k \geq 0$,

$$|f(2^{k\frac{4}{3}})| = W_\lambda(\frac{4}{3})E_{\kappa_\infty}(2^{k\frac{4}{3}}). \quad (79)$$

Consequently

$$\limsup_{x \rightarrow \infty} \frac{\log |f(x)| + (\log x)^2/(2a)}{\log x} = -\kappa_\infty. \quad (80)$$

Proof. Use (15) with $\Lambda = \lambda$ and (76):

$$\frac{|f(2^k(1+t))|}{E_{\kappa_\infty}(2^k(1+t))} \leq \lambda^{-1} \max_{1 \leq y \leq 2} W_\lambda(y).$$

At $t = 1/3$, $P_n(t) = \lambda^n$ exactly, which proves (79). Taking logarithms gives (80). $\square$

The exact ray (79) is sharp relative to its *actual* frequency. It is not the largest value in the shell whose left endpoint is 2^k : the fixed mantissa $4/3$ pays a power-sized displacement penalty. The true shell maximizer starts much closer to the left endpoint and spends a slowly growing transition time reaching the maximizing two-cycle. That is the next problem.

8 The sharp dyadic-annulus maximum and Lambert W

Define

$$M_k := \max_{2^k \leq x \leq 2^{k+1}} |f(x)|, \quad n := k + 1. \quad (81)$$

From (12),

$$M_k = \pi^{-n} 2^{-k(k+1)/2} A_n, \quad (82)$$

$$A_n := \max_{0 \leq t \leq 1} H(1+t)(1+t)^{-n} P_n(t). \quad (83)$$

All nontrivial lower-order behavior lies in A_n .

8.1 The boundary-layer coordinates

For $0 < t \leq 1/2$, write

$$t = 2^{-r} s, \quad r \in \mathbb{Z}_{\geq 0}, \quad \frac{1}{4} \leq s \leq \frac{1}{2}. \quad (84)$$

When $r < n$,

$$P_n(2^{-r} s) = \left(\prod_{h=1}^r \sin \frac{\pi s}{2^h} \right) P_{n-r}(s). \quad (85)$$

The small-angle block has the uniform factorization

$$\prod_{h=1}^r \sin \frac{\pi s}{2^h} = (\pi s)^r 2^{-r(r+1)/2} H_r(s), \quad H_r(s) := \prod_{h=1}^r \operatorname{sinc} \frac{\pi s}{2^h}. \quad (86)$$

On the compact interval $[1/4, 1/2]$, H_r is bounded above and below by positive constants uniformly in r .

The upper tail estimate follows from [theorem 7.2](#). A matching lower estimate, with enough freedom to optimize over s , comes from dense preimages of the maximizing cycle.

Lemma 8.1 (Near-maximizing preimages). *There is an absolute $C > 0$ such that, for every $s_0 \in [1/4, 1/2]$ and every integer $q \geq 1$, one can choose $s_q \in [1/4, 1/2]$ with*

$$|s_q - s_0| \leq 2^{-q}, \quad \mathbb{T}^q s_q \in \{1/3, 2/3\}, \quad (87)$$

and, for every $m \geq q$,

$$P_m(s_q) \geq \exp(m\beta - Cq^2). \quad (88)$$

Proof. The q fold preimages of $1/3$ are $(j + 1/3)/2^q$ and have spacing 2^{-q} ; using the preimages of $2/3$ near an endpoint gives (87). Before time q , every orbit point is a nonintegral rational with reduced denominator dividing $3 \cdot 2^{q-j}$. Since $|\sin \pi x| \geq 2 \operatorname{dist}(x, \mathbb{Z})$, the first q factors have logarithm bounded below by $-Cq^2$. Thereafter every factor is $\sqrt{3}/2 = e^\beta$. $\square$

Introduce the finite-dimensional objective

$$\mathcal{F}_n(r, s) := -\frac{a}{2}r^2 + r \left(\log(\pi s) - \frac{a}{2} - \beta \right) - n \log(1 + s2^{-r}). \quad (89)$$

Lemma 8.2 (Reduction to the boundary objective). *As $n \rightarrow \infty$,*

$$\log A_n = n\beta + \sup_{\substack{r \in \mathbb{Z}_{\geq 0} \\ 1/4 \leq s \leq 1/2}} \mathcal{F}_n(r, s) + \mathcal{O}((\log \log n)^2). \quad (90)$$

Proof. For the upper bound, the region $t \geq 1/2$ loses a fixed exponential factor through $(1+t)^{-n}$ and is below the eventual benchmark. In the region $0 < t < 1/2$, use (85), (86), and $P_{n-r}(s) \leq \lambda^{n-r-1}$. The factors $H(1+t)$ and $H_r(s)$ contribute only $\mathcal{O}(1)$. Values $r \geq n$ have all factors in the small-angle regime and are of order $\exp(-an^2/2 + \mathcal{O}(n))$, hence are negligible. This proves the claimed upper bound.

For the lower bound, take (r, s_0) within one of the supremum. Comparison with $r \asymp \log n$ shows that every such near-maximizer satisfies

$$r = \mathcal{O}(\log n), \quad ns_0 2^{-r} = \mathcal{O}((\log n)^2). \quad (91)$$

Choose $q = \lceil 4 \log_2 \log(n + e^e) \rceil$ and replace s_0 by the point s_q from theorem 8.1. On the localized set, the s derivative of $\mathcal{F}_n$ is $\mathcal{O}((\log n)^2)$, so this replacement changes the objective by $\mathcal{O}(1)$. The tail loses only $Cq^2 = \mathcal{O}((\log \log n)^2)$ in (88). All remaining factors are bounded below uniformly. This proves the matching lower bound. $\square$

8.2 The exact Lambert optimization

Put

$$C_0 := \log \pi - \frac{a}{2} - \beta = \log \left(\pi \sqrt{\frac{2}{3}} \right), \quad c := ae^{-C_0} = \frac{a}{\pi} \sqrt{\frac{3}{2}}. \quad (92)$$

Numerically, $c = 0.270222319733365 \dots$

Lemma 8.3 (Lambert- W saddle). *Let $w_n = W(cn)$, on the principal branch. Then*

$$\sup_{r, s} \mathcal{F}_n(r, s) = -\frac{w_n^2 + 2w_n}{2a} + \mathcal{O}(1), \quad (93)$$

where the supremum is over the set in (90).

Proof. At a maximizer, (91) implies that replacing $-n \log(1 + s2^{-r})$ by $-ns2^{-r}$ changes the value by $o(1)$. Let $A = C_0 + \log s$ and $z = ar - A$. A direct completion of the square gives

$$-\frac{a}{2}r^2 + r(C_0 + \log s) - ns2^{-r} = -\frac{z^2}{2a} - ne^{-C_0}e^{-z} + \frac{A^2}{2a}. \quad (94)$$

As s ranges over $[1/4, 1/2]$, A ranges over an interval of length exactly a . Successive integer values of r therefore make the corresponding z intervals meet end to end and tile the real line. The last term in (94) is uniformly bounded, so the supremum differs by $\mathcal{O}(1)$ from the real maximum of

$$-\frac{z^2}{2a} - ne^{-C_0}e^{-z}.$$

Its critical equation is $ze^z = ane^{-C_0} = cn$, hence $z = w_n$. At this point $ne^{-C_0}e^{-w_n} = w_n/a$, which gives (93). $\square$

The tiling is important: the continuous mantissa s , whose permitted multiplicative width is exactly two, absorbs the integer rounding in r . Without it one would lose an avoidable $\mathcal{O}(\log n)$ term.

Theorem 8.4 (Sharp dyadic-annulus envelope). *Let $n = k + 1$, $w_k = W(cn)$, and*

$$\gamma := \frac{a}{2} + \log \pi - \beta = \log \left(\frac{2^{3/2}\pi}{\sqrt{3}} \right). \quad (95)$$

Then

$$\log M_k = -\frac{a}{2}k(k+1) - n \log \pi + n\beta - \frac{w_k^2 + 2w_k}{2a} + \mathcal{O}((\log \log k)^2). \quad (96)$$

Equivalently,

$$\log M_k = -\frac{a}{2}k^2 - \gamma k - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2a} + \mathcal{O}((\log \log k)^2). \quad (97)$$

Proof. Combine (82), theorem 8.2, and theorem 8.3. The difference between the two displayed forms is a constant absorbed by the remainder. $\square$

Expanding $W(cn) = \log n - \log \log n + \log c + \mathcal{O}((\log \log n)/\log n)$ gives

$$\begin{aligned} \log M_k = & -\frac{a}{2}k^2 - \gamma k - \frac{(\log k)^2}{2a} + \frac{\log k \log \log k}{a} \\ & - \frac{1 + \log c}{a} \log k + \mathcal{O}((\log \log k)^2). \end{aligned} \quad (98)$$

In the physical scale $R = 2^k$, put $L = \log R$ and $d = \pi^{-1}\sqrt{3/2}$. Since $c(k+1) = d(L+a)$,

$$\begin{aligned} \log \max_{R \leq x \leq 2R} |f(x)| = & -\frac{L^2}{2a} - \kappa_\infty L \\ & - \frac{W(d(L+a))^2 + 2W(d(L+a))}{2a} + \mathcal{O}((\log \log L)^2). \end{aligned} \quad (99)$$

The first terms of the extra boundary loss are

$$-\frac{(\log \log R)^2}{2a} + \frac{\log \log R \log \log \log R}{a} + \mathcal{O}(\log \log R). \quad (100)$$

This is smaller than every fixed negative power of $\log R$, yet larger than every fixed negative power of R . It is the decisive correction missed by a normalizer containing only $(\log R)^{-p}$.

The proof also localizes a near-maximizer. At the smooth saddle,

$$t \asymp \frac{W(cn)}{an}, \quad r = \frac{W(cn)}{a} + \mathcal{O}(1) = \log_2 n - \log_2 \log n + \mathcal{O}(1). \quad (101)$$

Thus the largest shell peak lies at $x = 2^k(1 + \Theta(\log k/k))$, not at a fixed mantissa.

9 Exceptional two-adic troughs among the local peaks

Let

$$E_m := \max_{m \leq x \leq m+1} |f(x)| = |f(c_m)|, \quad (102)$$

where c_m is the unique critical point from [theorem 2.2](#). The sequence E_m has no single smooth asymptotic, even within one dyadic shell.

Fix an odd positive integer m , put $N = m2^k$, and write $x = N + t$ with $0 \leq t \leq 1$. Define

$$f_k(t) := \prod_{j=0}^k \operatorname{sinc}\left(\frac{\pi t}{2^j}\right). \quad (103)$$

For $0 \leq j \leq k$, $N/2^j$ is an integer and

$$\left| \operatorname{sinc}\left(\frac{\pi(N+t)}{2^j}\right) \right| = \frac{t}{N+t} \operatorname{sinc}\left(\frac{\pi t}{2^j}\right). \quad (104)$$

The remaining tail gives the exact identity

$$|f(N+t)| = \left(\frac{t}{N+t}\right)^{k+1} f_k(t) \left| H\left(m + \frac{t}{2^k}\right) \right|. \quad (105)$$

Because m is odd, $H(m) \neq 0$, and uniformly on $[0, 1]$,

$$f_k(t) = f(t)(1 + \mathcal{O}(4^{-k})), \quad H\left(m + \frac{t}{2^k}\right) = H(m)(1 + \mathcal{O}(2^{-k})). \quad (106)$$

Lemma 9.1 (A linear endpoint zero against a large power). *Let F be positive on $[0, 1]$ and $F(t) = h(1-t) + \mathcal{O}((1-t)^2)$ as $t \uparrow 1$, with $h > 0$. If $J_p = \max_{0 \leq t \leq 1} t^p F(t)$, then*

$$J_p \sim \frac{h}{ep}, \quad p(1 - t_p) \rightarrow 1 \quad (107)$$

for every maximizing point t_p .

Proof. Set $s = p(1-t)$. Uniformly for bounded s , $pt^p F(t) \rightarrow hse^{-s}$, whose unique maximum is h/e at $s = 1$. The bound $t^p F(t) \leq C(1-t)e^{-p(1-t)}$ near 1, together with exponential smallness away from 1, prevents a maximizing s from escaping to zero or infinity. $\square$

Since $(\operatorname{sinc}(\pi t))'|_{t=1} = -1$, formula (6) gives

$$f(t) = H(1)(1-t) + \mathcal{O}((1-t)^2), \text{ as } t \uparrow 1, \quad H(1) = 0.553771275888810\dots \quad (108)$$

Theorem 9.2 (Peaks following an odd multiple of a power of two). *Let m be fixed and odd, and write $c_{m2^k} = m2^k + t_k$. Then*

$$(k+1)(1-t_k) \rightarrow 1 \quad (109)$$

and

$$E_{m2^k} \sim \frac{H(1)|H(m)|}{e(k+1)} (m2^k)^{-(k+1)}. \quad (110)$$

In particular,

$$E_{2^k} \sim \frac{H(1)^2}{e(k+1)} 2^{-k(k+1)}. \quad (111)$$

Proof. Put $p = k + 1$. Equations (105) and (106) give, uniformly in t ,

$$|f(m2^k + t)| = (m2^k)^{-p} |H(m)| t^p f(t) (1 + o(1)).$$

Apply theorem 9.1 and (108). The uniform multiplicative error preserves both the value and location asymptotics. $\square$

Comparing (111) with (97) gives

$$\log \frac{E_{2^k}}{M_k} = -\frac{a}{2} k^2 + \mathcal{O}(k). \quad (112)$$

Thus peaks whose locations differ by less than a factor two can differ by an additional log-normal factor. This decisively rules out a smooth asymptotic curve for all interzero maxima.

10 Comparative audit of the supplied materials

The reports are best viewed as studying different functionals of the same finite product P_n , not as proposing mutually exclusive pointwise formulas. The following table records the substantive contribution and the correction needed for each source.

Material	Correct contribution	Required correction or scope restriction
Original MSE source [9]	Correctly asks an averaged question and already notices integer zeros, Thue–Morse signs, and a log-normal-looking scale.	It does not specify which mean of the lacunary numerator determines the power correction; the supplied answer does not solve Q2 or Q3.
First report [14]	Exact product and lobe structure; typical exponent; exact martingale coboundary; CLT and the corrected LIL variance; sharp annular expansion through $\mathcal{O}(\log k)$.	It does not solve the L^1 /Cesàro problem, and its annular error is weaker than the later Lambert- W result.
Second report [15]	Strongest treatment of the dyadic-annulus maximum: exact Lambert- W formula with $\mathcal{O}((\log \log k)^2)$ error; dense-preimage construction; exact two-adic peak troughs.	It explicitly recognizes that the original Cesàro question is a separate L^1 problem and therefore does not answer it.
Third report [16]	Exact dyadic factorization; typical, L^1 , L^2 , and L^∞ exponents; Q3; dyadic endpoint normalization; singular dyadic profile.	It imports the coefficient π from Aistleitner–Hofer–Larcher. The correct coefficient is $\pi/\sqrt{2}$ relative to $\sqrt{n \log \log n}$.
Fourth report [17]	Most complete transfer-operator spectrum; exact L^2 identity; explicit Collatz–Wielandt certificate for ρ_1 ; Q3 and phase profile; sharp uniform envelope; unique lobes and exceptional two-adic peaks.	Its uniform envelope is a fixed-ray/global-limsup statement, not the finer maximum relative to the left endpoint of each dyadic shell. The Lambert boundary layer from the second report is still needed.
AHL lacunary-product source [1]	Supplies the relevant lacunary-product framework and a rigorous enclosure for the L^1 growth constant.	Its displayed f_1 variance omits the cosine Parseval factor $1/2$, doubling the variance and multiplying the LIL coefficient by $\sqrt{2}$.
Comparative audit [18]	Correctly unifies the L^q spectrum, recognizes the LIL variance error, derives Q3 and the singular phase obstruction, and adds logarithmic averaging.	“Unattainable in principle” is justified only for gauges with stabilized within-shell shape. Its quoted numerical phase span comes from a coarse mantissa grid and misses narrower features of the singular-continuous profile.

10.1 The two apparent contradictions that are not contradictions

Fixed maximizing ray versus shell maximum. Along $x = 2^k(4/3)$, the ratio $|f(x)|/E_{\kappa_\infty}(x)$ is exactly constant. In the shell $[2^k, 2^{k+1}]$, however, comparing with the scale at the left endpoint 2^k charges the fixed ray the factor $E_{\kappa_\infty}(4 \cdot 2^k/3)/E_{\kappa_\infty}(2^k)$, which is a negative power of 2^k . A boundary-layer point much closer to 2^k is larger even though it has not followed the maximizing orbit from time zero. Hence the exact fixed-ray theorem and the extra Lambert suppression in the shell maximum are simultaneously true.

The RMS exponent versus the arithmetic exponent. The identity $\int P_n^2 = 2^{-n}$ makes $1/\sqrt{2}$ the exact RMS numerator rate. The arithmetic numerator rate is the different Perron root $\rho = 0.6613226 \dots$. Since $\rho < 1/\sqrt{2}$, the arithmetic-normalized amplitude decays with the larger power $\kappa_1 > \kappa_2$. Treating an L^2 computation as a pointwise or L^1 computation is exactly the source of the early exponent mismatch.

10.2 What is now final, and what remains genuinely open

The following statements are settled by the arguments assembled here.

1. The exact dyadic reduction and all four first-order exponents.
2. The typical CLT and corrected two-sided LIL.
3. Q3 for an explicit elementary analytic gauge and Q2 at dyadic endpoints.
4. Failure of full Q2 for that gauge and for every stabilized within-shell gauge.
5. The sharp fixed-ray/global envelope and the sharper dyadic-annulus Lambert- W asymptotic.
6. The exceptional two-adic asymptotic for local peaks.

The only normalization issue not settled by the supplied methods is the bare existence of a deliberately non-self-similar, scale-dependent, positive analytic monotone gauge satisfying ordinary Q2. Such a gauge, if constructed, would have to encode increasingly fine information about the singular conformal distribution. It would not be an elementary asymptotic decay law in the usual sense.

11 Independent numerical checks and reproducibility

Numerics are not used to prove any exponent. They are useful for detecting normalization mistakes and for checking that the constants attached to the proved formulas are consistent.

11.1 Direct dyadic-cell quadrature

Every zero of P_n lies on the grid $2^{-n}\mathbb{Z}$. On each of the 2^n finest dyadic cells the absolute product is smooth, so fixed-order Gauss–Legendre quadrature is highly effective. Let

$$I_n := \int_0^1 P_n(t) dt, \quad \mathcal{A}_n := \rho^{-n} \int_0^1 W_1(1+t) P_n(t) dt. \quad (113)$$

The computations below used eight Gauss points per finest cell and sixty tail factors in H ; changing to six, twelve, or sixteen points leaves the displayed digits stable.

n	I_n/I_{n-1}	$\mathcal{A}_n$
8	0.661322173532878	0.091266135409708
10	0.661322575197187	0.091266124753463
12	0.661322601094757	0.091266124150951
14	0.661322602051094	0.091266124131594
16	0.661322602062189	0.091266124131543

This independently reproduces (49) and (57). The analogous logarithmic-shell sequence converges to

$$B_1 = 0.065059235040389 \dots, \quad B_1/a = 0.093860635756799 \dots \quad (114)$$

A minimal implementation is:

```
import numpy as np
from numpy.polynomial.legendre import leggauss

rho = 0.661322602060565
loga = np.log(2.0)
kappa = 0.5 + np.log(np.pi / rho) / loga

def H(y, terms=60):
    ans = np.ones_like(y)
    for m in range(1, terms + 1):
        ans *= np.sinc(y / 2**m) # np.sinc(u)=sin(pi*u)/(pi*u)
    return np.abs(ans)

def shell_data(n, order=8):
    xi, wi = leggauss(order)
    total_I = 0.0
    total_A = 0.0
    cells = 2**n
    for j in range(cells):
        t = (j + (xi + 1.0)/2.0) / cells
        w = wi / (2.0*cells)
        P = np.ones_like(t)
        for ell in range(n):
            P *= np.abs(np.sin(np.pi * 2**ell * t))
        y = 1.0 + t
        W = (H(y) * (rho/np.pi) * y**(kappa-1.0)
              * np.exp(np.log(y)**2/(2.0*loga)))
        total_I += np.dot(w, P)
        total_A += np.dot(w, W*P)
    return total_I, total_A/rho**n
```

11.2 The phase profile has narrow singular features

At finite level define

$$\mathcal{F}_n(z) := \frac{\int_0^z W_1(1+t)P_n(t) dt}{\int_0^1 W_1(1+t)P_n(t) dt}, \quad \mathcal{P}_n(1+z) := \frac{1 + \mathcal{F}_n(z)}{1+z}. \quad (115)$$

For $n = 13$, the same dyadic-cell quadrature gives

y	$\mathcal{P}_{13}(y)$	comment
8/7	0.9163240341	narrow lower feature
1.150	0.9248812037	point seen on a grid of spacing 0.025
10/7	1.1181043233	narrow upper feature
1.425	1.1130801816	point seen on a grid of spacing 0.025

At $n = 17$, the values at $8/7$ and $10/7$ are approximately 0.9162741 and 1.1181341. These computations explain the discrepancy with the comparative audit's coarser quoted range. They are not certified global minimum and maximum values: the limit is a singular-continuous distribution, and ever finer features are expected.

11.3 Finite-shell check of the Lambert formula

Brute-force enumeration of the 2^n smooth lobe cells, followed by local one-dimensional maximization, gives the following comparison. The column "Lambert core" is the right side of (97) without its remainder.

k	maximizing t in $x = 2^k(1+t)$	$\log M_k$	$\log M_k$ —Lambert core
8	0.16637624	−39.27401113	−2.00423
12	0.16664866	−74.15774249	−2.07119
16	0.08333446	−119.95217544	−2.04250

The bounded-looking residual at these modest levels is fully compatible with the proved $\mathcal{O}((\log \log k)^2)$ error. The jumps in the maximizing mantissa reflect changes among preimages of the maximizing two-cycle.

11.4 An exact-arithmetic certificate for a coarse Perron-root interval

For completeness, the fourth report's Collatz–Wielandt certificate can be checked with exact rational arithmetic. Take $h(x) = F(\sin \pi x)$, where

$$F(u) = 1 + \frac{376189}{10^6}u - \frac{69093}{10^6}u^2 + \frac{15483}{10^6}u^3.$$

The following Sturm-sequence check proves $0.66126807 < \rho < 0.66134891$ without floating-point assumptions.

```
import sympy as sp

t = sp.symbols("t", real=True)
u = 2*t/(1+t**2)
v = (1-t**2)/(1+t**2)

c = sp.Rational(376189, 10**6)
d = sp.Rational(-69093, 10**6)
e = sp.Rational(15483, 10**6)
F = lambda z: 1 + c*z + d*z**2 + e*z**3

R = sp.cancel((u*F(u) + v*F(v)) / (2*F(2*u*v)))
lo = sp.Rational(66126807, 10**8)
hi = sp.Rational(66134891, 10**8)

for expr in (R-lo, hi-R):
    num, den = map(lambda p: sp.Poly(p, t),
                    sp.together(expr).as_numer_denom())
    assert num.eval(0) > 0 and den.eval(0) > 0
    assert num.count_roots(0, 1) == 0
    assert den.count_roots(0, 1) == 0
```

The substitution $t = \tan(\pi x/4)$ converts $(\mathcal{L}_1 h)(x)/h(x)$ into the rational function $R(t)$. The two inequalities then follow from Collatz–Wielandt.

12 Conclusion

The final answer is not one corrected equivalent. It is a hierarchy generated by the exact identity (15).

The deterministic denominator contributes $-(\log x)^2/(2 \log 2)$. The lacunary numerator then chooses the power of x : $1/2$ for the geometric mean and almost-everywhere ray, ρ_1 for the absolute mean, $1/\sqrt{2}$ for RMS, and $\sqrt{3}/2$ for the fixed-ray supremum. The typical logarithm has genuine Gaussian and LIL fluctuations, with corrected variance $\pi^2/4$. The ordinary Cesàro question is governed by the arithmetic exponent κ_1 , not by the RMS exponent. Its weaker bounded form has the explicit analytic solution $A_1 E_{\kappa_1}$; the all- t limit for that entire natural class is obstructed by a singular dyadic phase. The logarithmic mean removes that phase.

At the extremal end, the maximizing period-two orbit proves the sharp global power exponent, while the largest value relative to a dyadic shell's left endpoint is a different boundary-layer problem. Its transition time is selected by Lambert W and creates a second log-normal correction in $\log \log x$. Finally, the local peak sequence retains the exact two-adic multiplicity structure of the integer zeros, producing troughs far below the annular maximum.

For a future Lean formalization, the development naturally separates into independent blocks: locally uniform product convergence and dilation; the canonical zero product and strict log-concavity of lobes; exact dyadic factorization; the finite L^2 identity; the elementary two-step Gelfond inequality; rational preimages and Lambert optimization; the boundary maximum lemma; and, as imported analytic interfaces, Birkhoff/martingale limit theory and the spectral theorem for $\mathcal{L}_q$. No ambiguity about "the decay rate" remains once the functional being normalized is stated.

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